

GIRIJANANDA CHOWDHURY UNIVERSITY, AZARA, GUWAHATI 781017, ASSAM
DEPARTMENT OF CIVIL ENGINEERING
Formative Assessment -II

Course: BCE23404T (Hydraulic Engineering)
Date: 22/04/2025
Duration: 1 hour

Academic Year: 2024-25
Total Marks: 30

Question		Marks	Course Outcome (s)
1	For the velocity profile for laminar boundary flow given as: $\frac{u}{U} = \frac{3}{2} \left(\frac{y}{\delta} \right) - \frac{1}{2} \left(\frac{y}{\delta} \right)^3$ Find an expression for boundary layer thickness, shear stress and co-efficient of drag in terms of Reynolds number.	15	CO3
2	What do you understand by the terms: Major energy loss and minor energy losses in pipes?	8	CO2
3.	Write a short notes on: i. Displacement thickness ii. Momentum thickness	7	CO2